8.19 Accumulator Component, ACCUM

An accumulator component is indicated by ACCUM for Word 2 on card CCC0000. For major edits, minor edits, and plot variables, the volume in the accumulator component is numbered CCC010000, and the junction in the accumulator component is numbered CCC010000.

An accumulator is a lumped parameter component treated by special numerical techniques that model both the tank and surgeline until the accumulator is emptied of liquid. When the last of the liquid leaves the accumulator, the code automatically resets the accumulator to an equivalent single-volume with an outlet junction and proceeds with calculations using the normal hydrodynamic solution algorithm.

In the following input requirements, it is assumed that the component is an accumulator in which liquid completely fills the surge line but may or may not occupy the tank. It is further assumed that the accumulator is not initially in the injection mode. Hence, the initial pressure must be input lower than the injection point pressure, including elevation head effects. The junction initial conditions may not be input i.e., initial liquid and gas velocities are set to 0 in the code. The noncondensable gas in the accumulator is nitrogen and that the gas and liquid are initially in equilibrium. No other junctions should be connected to the accumulator volume. The geometry of the tank may be cylindrical or spherical. The standpipe/surge line inlet refers to the end of the pipe inside the tank itself.

8.19.1 Accumulator Volume Geometry, CCC0101-0109

This card is required.

W1(R) Area (m^2, ft^2). This is the flow area of a cylindrical tank, or the maximum flow area of a spherical tank. In the case of a spherical tank, the flow area and the tank radius are related by the formula $A = \pi R^2$.

W2(R) Length (m, ft). This is the length of the tank above the standpipe/surgeline inlet, where this inlet refers to the end of the pipe inside the tank itself.

W3(R) Volume (m^3, ft^3). This is the volume of the tank above the standpipe/surgeline inlet, where this inlet refers to the end of the pipe inside the tank itself. The code requires that the volume, area, and length are consistent. For a cylindrical tank, W3 = W1•W2, and at least two of the three quantities, W1, W2 or W3, must be nonzero. If one of the quantities is 0, it will be computed from the other two. For a spherical tank, W1 and W2 must be nonzero. If W3 is 0, it will be computed from the other two. If none of the words are 0, they must satisfy the consistency condition within a relative error $\pm$ 0.000001.

W4(R) Azimuthal angle (degrees). The absolute value of this angle must be $\leq$ 360 degrees. This quantity is not used in the calculation but is specified for possible automated drawing of nodalization diagrams.

W5(R) Inclination angle (degrees). Only +90 or -90 degrees is allowed. The accumulator is assumed to be a vertical tank with the standpipe/surgeline inlet (where this inlet refers to the end of the pipe inside the tank itself) at the bottom. This angle is used in the interphase drag calculation.

W6(R) Elevation change (m, ft). This is the elevation change from the standpipe/surgeline inlet (where this inlet refers to the end of the pipe inside the tank itself) to the top of the tank. A positive value is an increase in elevation. The absolute value of this quantity must be nonzero, less than or equal to the length, and have the same sign as the angle for vertical orientation. As with other components, this Word 6 is compared to the length (Word 2) to decide if the horizontal or vertical flow regime map is used. This is not important for this component, since the correlations that depend on the flow regime map are not needed for this component. The volume conditions are determined from the accumulator's special model.

W7(R) Wall roughness (m, ft).

W8(R) Hydraulic diameter (m, ft). This should be computed from $4.0 \bullet \left(\frac{\text{area}}{\text{wetted perimeter}}\right)$.

If 0, the hydraulic diameter of the tank is computed from $2.0 \bullet \left(\frac{\text{area}}{\pi}\right)^{0.5}$. A check is made that the pipe roughness is less than half the hydraulic diameter of the tank. See Word 1 for the area.

W9(I) Volume control flags. This word has the format tlpvbfe. It is not necessary to input leading zeros. Volume flags consist of scalar oriented and coordinate direction oriented flags. Only one value for a scalar oriented flag is entered per volume but up to three coordinate oriented flags can be entered for a volume, one for each coordinate direction. At present, the f flag is the only coordinate direction oriented flag. This word enters the scalar oriented flags and the x-coordinate flag. The accumulator component forces all volume flags except for the x-coordinate f digit, and y- and z-coordinate flags are not read. For the accumulator volume control flag, the format is restricted to 00110f0.

f-flag The digit f specifies whether wall friction is to be computed. f = 0 specifies the wall friction effects are to be computed along the x-coordinate; and f = 1 specifies the friction effects are not to be computed along the x-coordinate.

W10(I) Geometry flag (optional). To specify a cylindrical tank, set the flag equal to 0 (default). To specify a spherical tank, set the flag equal to 1. To specify a cylindrical tank with fluidic device, set the flag equal to 2.

8.19.2 Accumulator Additional Laminar Wall Friction, CCC0131

This card is optional. and if this card is not entered, the default values are 1.0 for the laminar shape factor and 0.0 for the viscosity ratio exponent.

W1(R) Shape factor for x-coordinate.

W2(R) Viscosity ratio exponent for x-coordinate.

8.19.3 Accumulator Alternate Wall Friction Data, CCC0141

This card is optional. The input on this card, which does no distinguish between forward and reverse loss coefficients, has been superseded by the CCC1101-1102 cards. If this card is entered, the form loss coefficient is calculated from

$$K = A + BRe^{-C}$$

where K is the form loss coefficients. Re is the Reynolds number based on mixture fluid properties. If this card is being used for the form loss calculation, then enter all three words.

W1(R) A (≥ 0). This quantity must be greater than or equal to 0.

W2(R) B (≥ 0). This quantity must be greater than or equal to 0.

W3(R) C (≥ 0). This quantity must be greater than or equal to 0.

8.19.4 Accumulator Tank Initial Thermodynamics Conditions, CCC0200

This card is required.

W1(R) Pressure (Pa, lb_f/in^2).

W2(R) Temperature (K, ^oF).

W3(R) Boron concentration (mass of boron per mass of liquid). This word is optional.

8.19.5 Accumulator Junction Geometry, CCC1101

This card is required.

W1(I) <u>To</u> connection code to a component. The <u>from</u> connection is not entered, since it is always from the accumulator. The <u>to</u> connection code refers to the component from which the junction coordinate direction originates. The connection code is CCCVV000N, where

CCC is the component number, VV is the volume number, and N indicates the face number. A nonzero N specifies the expanded format. The number N equal to 1 and 2 specifies the inlet and outlet faces respectively for the volume's coordinate direction. The number N equal to 3-6 specifies crossflow. The number N equal to 3 and 4 would specify inlet and outlet faces for the second coordinate direction; N equal to 5 and 6 would do the same for the third coordinate direction. For connecting to a time-dependent volume, N is restricted to 1 or 2.

W2(R) Area (m^2, ft). This is the average area of the surgeline and standpipe.

W3(R) Reynolds number independent forward flow energy loss coefficient, A_F. This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is positive or 0. Because the smooth area change model (control flag a = 0, see W5) must be used for the junction in the ACCUM component, the loss coefficient entered here should be consistent with the junction flow area entered in W2.

W4(R) Reynolds number independent reverse flow energy loss coefficient, A_R. This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is negative. See information for the previous word regarding interpretation and use of this loss coefficient.

W5(I) Junction control flags. This word has the format jefvcahs. It is not necessary to input leading zeros. For the accumulator, the junction control flag is restricted to the format 0000c0hs.

The accumulator model automatically disables the following flags as long as liquid remains in the accumulator. However, when the accumulator empties of liquid, the model is automatically converted to an active normal volume. The following flags are then enabled and used as defined.

c-flag The digit c specifies choking options. c = 0 means that the choking model will be applied, and c = 1 means the choking model will not be applied. The choking model used is based on input on card 1. If card 1 is missing or it does not contain Option 50, the Henry-Fauske critical flow model is active. If card 1 is present and it contains Option 50, the original RELAP5 critical flow model is active. Unlike junctions for certain other hydrodynamic components, only the default values of discharge coefficients and factors used in the critical flow models are allowed for the junction in the ACCUM component. Problem renodalization should be used to circumvent this limitation should it prove to be significant. For the original RELAP5 critical flow model, the default values for the subcooled, two-phase and superheated discharge coefficients each are 1.0. For the Henry-Fauske critical flow model, the default value for the discharge coefficient is 1.0 and the default value for the thermal nonequilibrium constant is 0.14.

h-flag The digit h specifies nonhomogeneous or homogeneous. h = 0 specifies the nonhomogeneous (two-velocity momentum equations) option; h = 1 or 2 specifies the homogeneous (single-velocity momentum equation) option. For the homogeneous option (h = 1 or 2), the major edit will show h = 1.

s-flag The digit s specifies momentum flux options. s = 0 uses momentum flux in both the to volume and the from volume. s = 1 uses momentum flux in the from volume, but not in the to volume. s = 2 or 3 is not allowed for an accumulator.

8.19.6 Accumulator Junction Form Loss Data, CCC1102

This card is optional. The user-specified form loss is given in Words 3 and 4 of card CCC1101 if this card is not entered. If this card is entered, the form loss coefficient is calculated from

$$K_F = A_F + B_F Re^{-C_F}$$

$$K_R = A_R + B_R Re^{-C_R}$$

where K_F and K_R are the forward and reverse form loss coefficients. A_F and A_R are the Words 3 and 4 of card CCC1101. Re is the Reynolds number based on mixture fluid properties. If this card is being used for the form loss calculation, then enter all four words for the appropriate expression.

W1(R) B_F (≥ 0). This quantity must be greater than or equal to 0.

W2(R) C_F (≥ 0). This quantity must be greater than or equal to 0.

W3(R) B_R (≥ 0). This quantity must be greater than or equal to 0.

W4(R) C_R (≥ 0). This quantity must be greater than or equal to 0.

8.19.7 Accumulator Miscellaneous Parameters, CCC2200

This card is required.

W1(R) Liquid volume in tank (m^3, ft^3). This is the volume of water contained in the tank above the standpipe/surgeline inlet (where this inlet refers to the end of the pipe inside the tank).

W2(R) Liquid level in tank (m, ft). This is the liquid level of water contained in the tank above the standpipe/surgeline inlet (where this inlet refers to the end of the pipe inside the tank) entrance. For a cylindrical tank, either W1 or W2 must be specified as nonzero. For a spherical tank, W2 must be specified as nonzero. If one of the words is 0, it is computed from the other two.

W3(R) Length of surgeline and standpipe (m, ft). If input as 0, then the surgeline and standpipe are not modeled.

W4(R) Elevation drop of surgeline and standpipe (m, ft). This is the elevation drop from the standpipe/surgeline inlet (where this inlet refers to the end of the pipe inside the tank) entrance to the injection point. A positive number denotes a decrease in elevation.

W5(R) Tank wall thickness (m, ft). This is not allowed to be 0.

W6(I) Heat transfer flag. If 0, heat transfer will be calculated. If 1, no heat transfer will be calculated.

W7(R) Tank density (kg/m^3, lb/ft^3). If 0, the density will default to that for carbon steel.

W8(R) Tank volumetric heat capacity (J/kg•K, Btu/lb•^{o}F). If 0, the heat capacity will default to that for carbon steel.

W9(I) Trip number. If 0 or if no number is input, then no trip test is performed. If nonzero then this must be a valid trip number, the operations performed are similar to those performed for a trip valve. If the trip is false, then the accumulator is isolated and no flow through the junction can occur. If the trip is true, then the accumulator is not isolated and flow through the junction will occur in the normal manner for an accumulator.

8.19.8 Fluidic Device Geometry, CCC2201

This cars is required if the geometry flag(W10) of accumulator geometry card CCC0101-0109 is set to 2 (cylindrical tank with fluidic device).

W1(R) Length of the stand pipe of fluidic device (m, ft). This value shlould be less than the length of tank.

W2(R) Reynolds number independent forward energy loss coefficient due to vortex motion. This quantity will be added to junction loss coefficient.

W3(R) Reynolds number independent reverse energy loss coefficient due to vortex motion. This quantity will be added to junction loss coefficient.

8.20 Sub-Domain Boundary Volume (SDBVOL) Component

A sub-domain boundary volume represents the three-dimensional cell that is connected to a one-dimensional cell.